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Date: 1/15/24

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Course: Calculus Beta

Assignment: MyLab Math - Uge 13

Find ∇f at the given point.

$$f(x,y,z) = x^2 + y^2 - 5z^2 + z \ln x, \quad (1,3,2)$$

The gradient vector (gradient) of $f(x,y,z)$ at a point $P_0(x_0, y_0, z_0)$ is the vector

$$\nabla f = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$$

obtained by evaluating the partial derivatives of f at P_0 .

Find $\frac{\partial f}{\partial x}$.

$$\frac{\partial f}{\partial x} = 2x + \frac{z}{x}$$

Find $\frac{\partial f}{\partial y}$.

$$\frac{\partial f}{\partial y} = 2y$$

Find $\frac{\partial f}{\partial z}$.

$$\frac{\partial f}{\partial z} = -10z + \ln x$$

Now, find the value of $\frac{\partial f}{\partial x}$ at $(1,3,2)$.

$$\begin{aligned} \left. \frac{\partial f}{\partial x} \right|_{(1,3,2)} &= \left(2x + \frac{z}{x} \right) \Big|_{(1,3,2)} \\ &= 2(1) + \frac{2}{1} \\ &= 4 \end{aligned}$$

Now, find the value of $\frac{\partial f}{\partial y}$ at $(1,3,2)$.

$$\begin{aligned} \left. \frac{\partial f}{\partial y} \right|_{(1,3,2)} &= 2y \Big|_{(1,3,2)} \\ &= 2(3) \\ &= 6 \end{aligned}$$

Now, find the value of $\frac{\partial f}{\partial z}$ at $(1,3,2)$.

$$\begin{aligned} \left. \frac{\partial f}{\partial z} \right|_{(1,3,2)} &= (-10z + \ln x) \Big|_{(1,3,2)} \\ &= -10(2) + \ln 1 \\ &= -20 \end{aligned}$$

Therefore, ∇f at $(1,3,2)$ is $\nabla f \Big|_{(1,3,2)} = 4\mathbf{i} + 6\mathbf{j} - 20\mathbf{k}$.